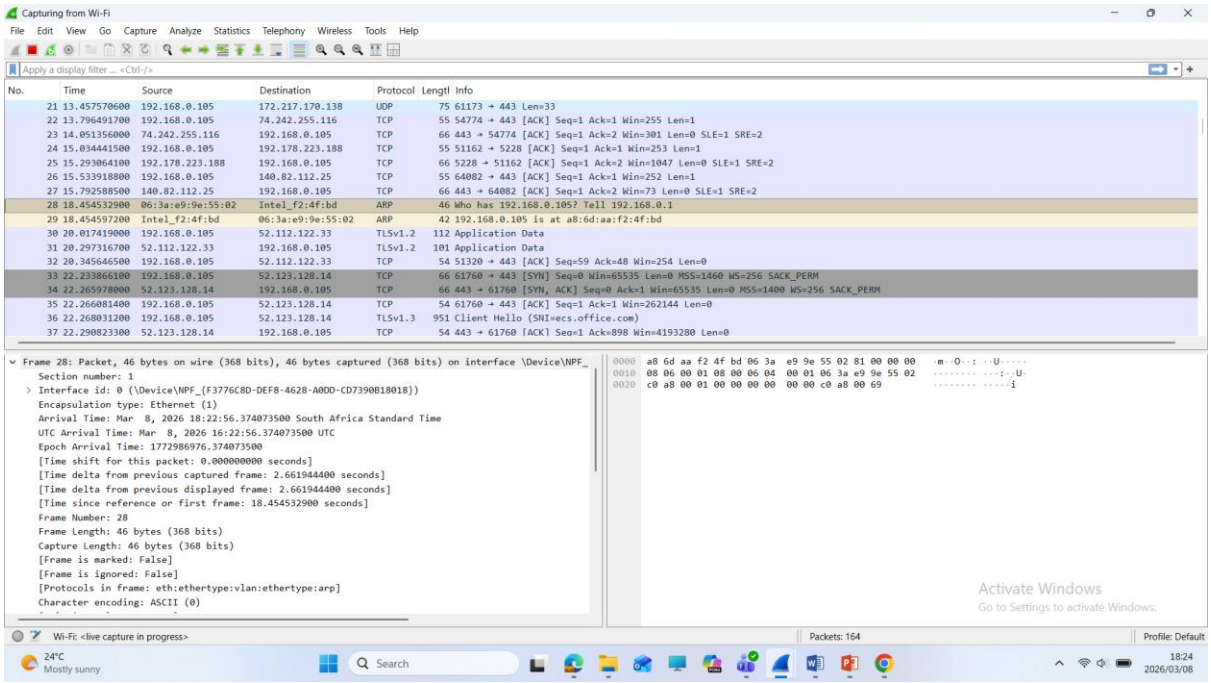
Protocol	OSI Layer	TPC/IP Layer	Purpose
Ethernet	Layer 2-DATA LINK	Network ACCESS	Connects to devices in the LAN to share data an communication like MAC
ARP (Address Resolution Protocol)	Layer 2-DATA LINK	Network ACCESS	Dynamic IP address into the MAC address on the LAN
IP(Internet Protocol)	Layer 3-NETWORK	Internet	Addresses routing and data packet
ICMP(Internet Control Message Protocol)	Layer 3-NETWORK	Internet	Sends errors and operational information (e.g traceroute)
TCP/UDP	Layer 4 -TRANSPORT	Transport	Manages host to host communication. TCP offers more reliability and the UDP is faster an more efficient
DNS,HTTP,HTTPS	Layer 7- APPLICATION	Application	Secure web browsing and communication ,readable domain names

2. Packet Encapsulation in Wireshark

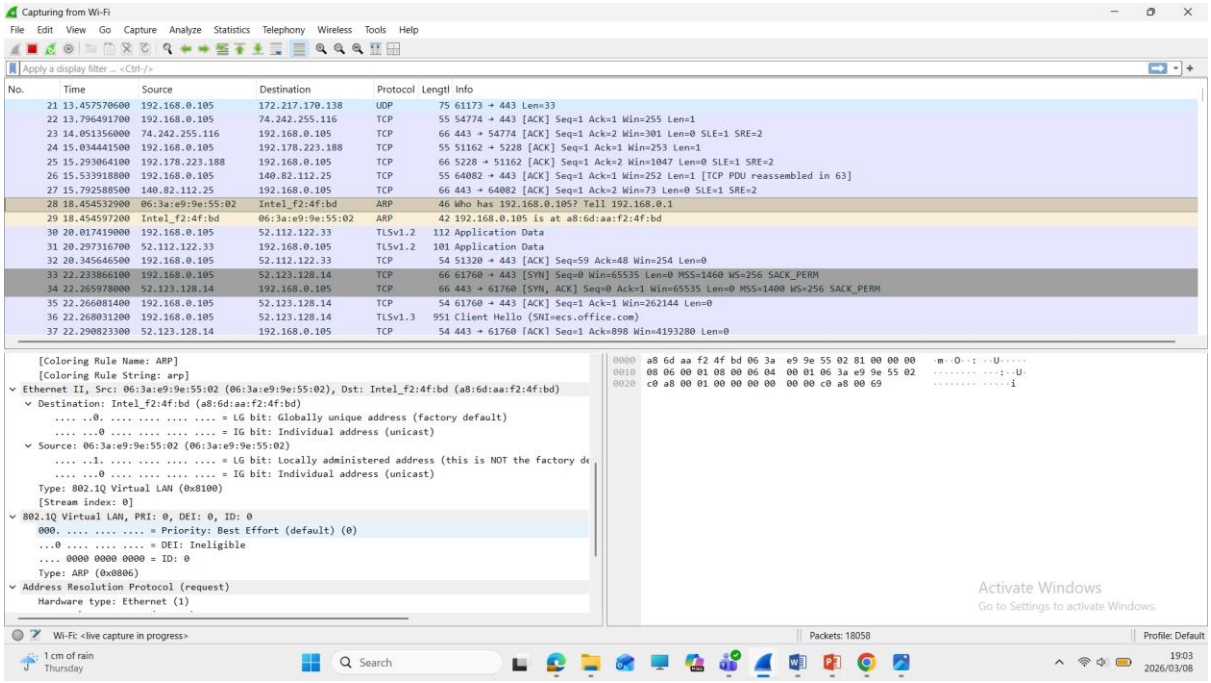
Frame



The ARP is being sent to a local network which is located in Layer 2 in the OSI Layer and is the Network Access in the TPC/IP. The ARP IP address is **192.168.0.1** and is asking **Who has** its destination IP address is **192.168.0.105** . Before it sends it is going to need the MAC (Medium Access

Control) Address to see weather the local IP address is responsive to go through. This Frame size is 46bytes (368 bits) it also the time of arrival for both IP Address

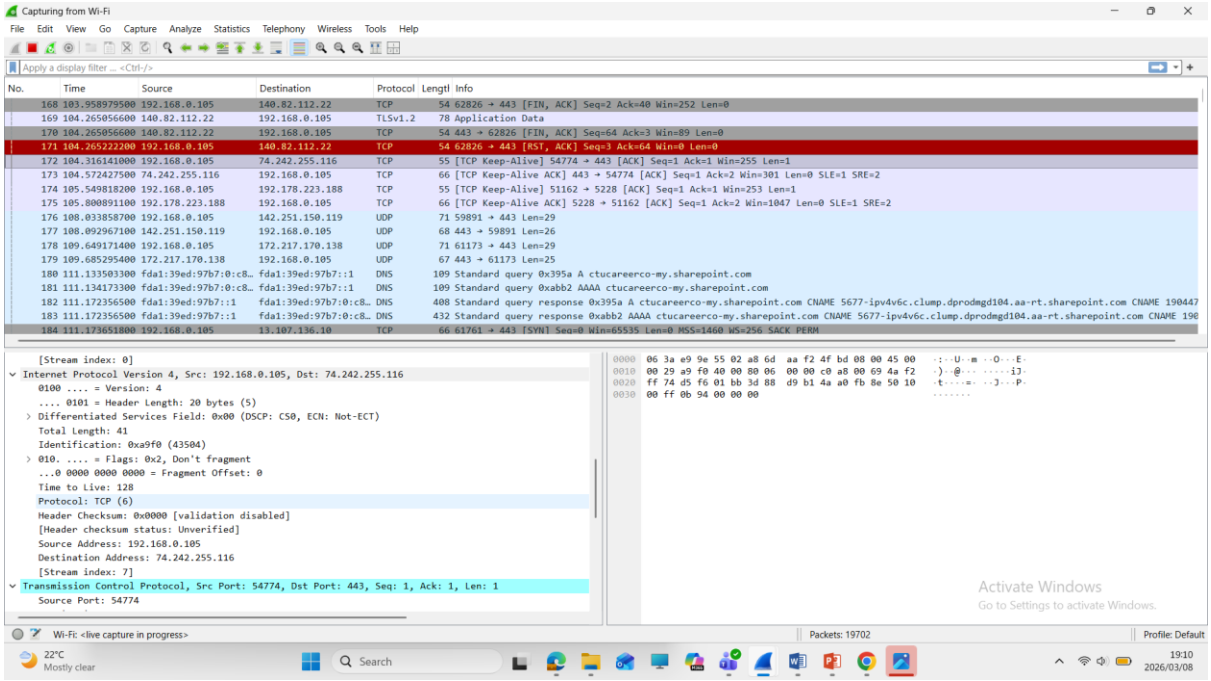
Ethernet



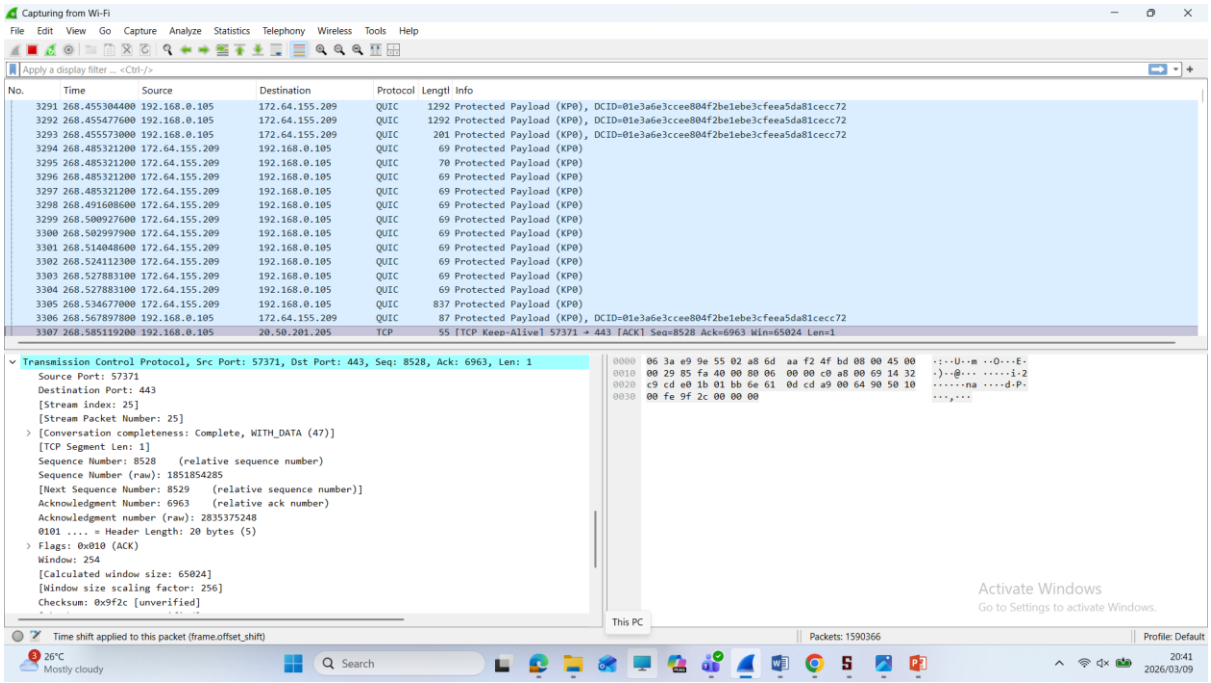
In this picture we also see the MAC Address and where it destination is , **`a8:6d:aa:f2:4f:bd` (Intel network adapter)** is the destination for the receiver . The indication of **802.1Q VLAN tag** means that the packet is being transmitted within the virtual LAN before the next protocol is processed. In this frame is where the ARP is requesting the MAC address to communicate within the local address . Once the receiver responses MAC Address the sender will update its ARP

IPV4

In this image we see that the packet is being sent from the receiver to the external server **74.242.255.116**. The IPV4 shows the header length which is 20 bytes and that it is carrying a TCP segment. The receiver **(192.168.0.105)** is communicating with the remote server **TCP port 443 (HTTPS)** . The highlighted packet shows that the device is forcefully resetting or terminating the connection which ends up closing all communication with the remote sever



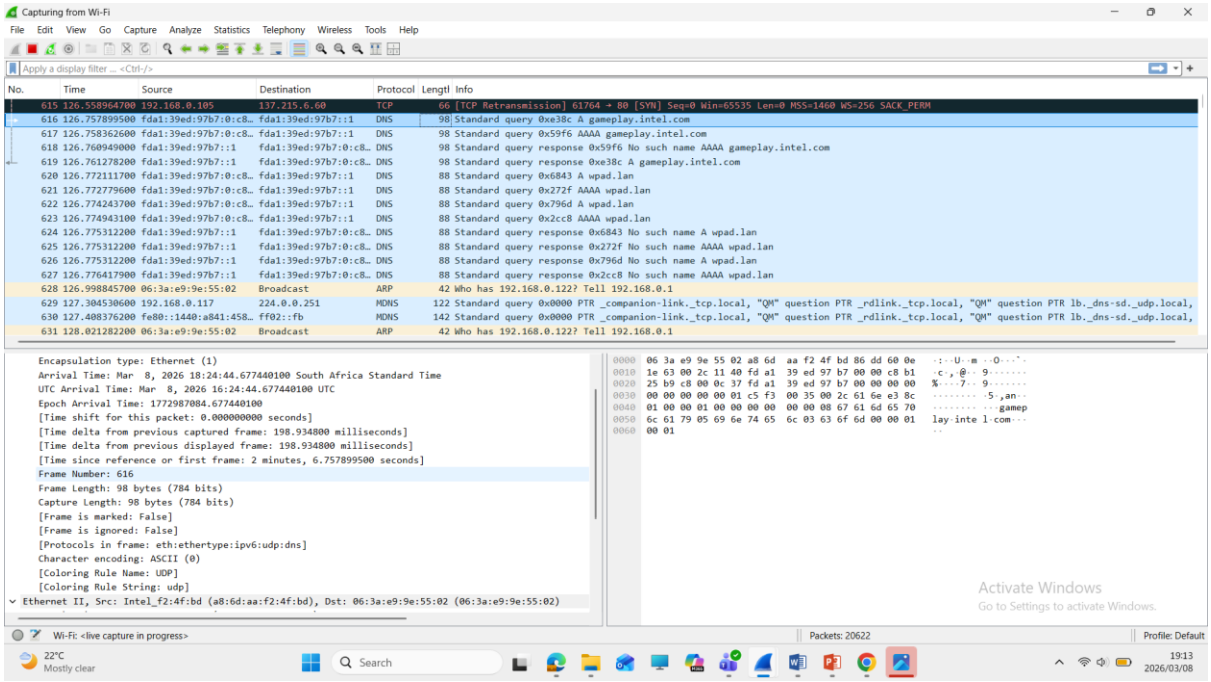
TCP/UDP



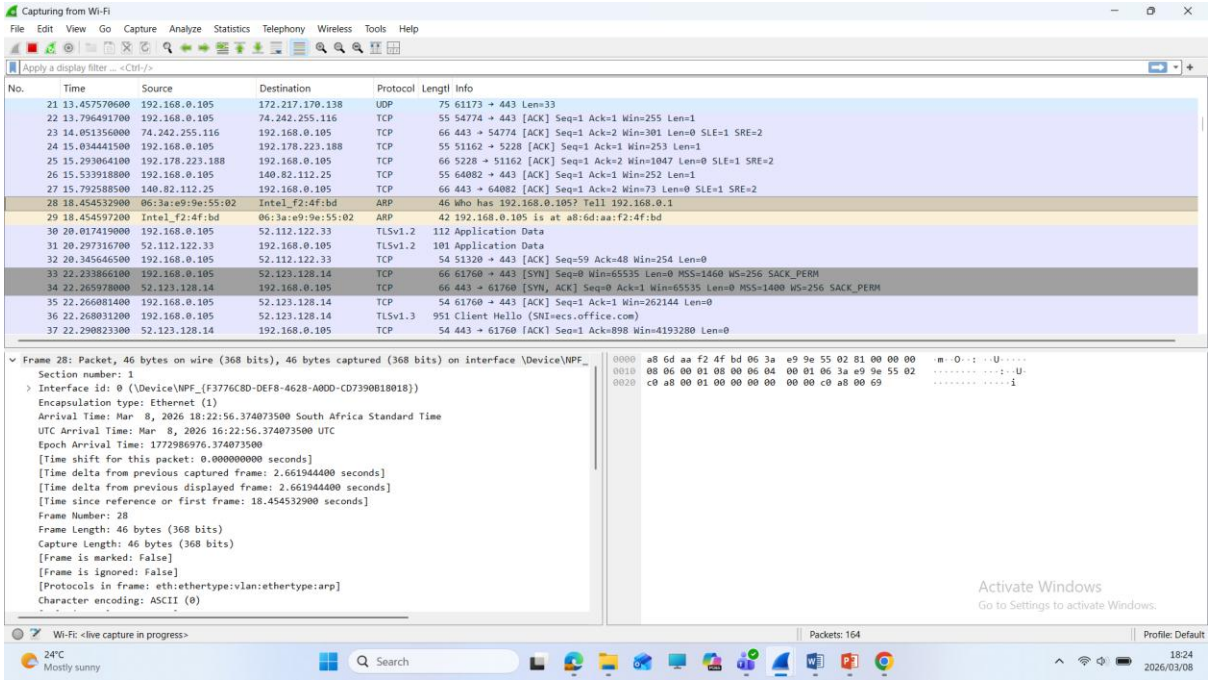
In this image we see that the TCP communication with the receiver **192.168.0.105** . We see that the TCP is using **source port 57371** to a remote sever and its destination sever port is **443 (HTTPS)** , this image shows that the packet has successfully been received . This packet length is very small (**Len = 1**) and is marked to keep the TCP alive so it can maintain the connection to the remote sever . The **UDP** which you would find in the **QUIC packets** above

3. Observe Specific Protocol in Wireshark

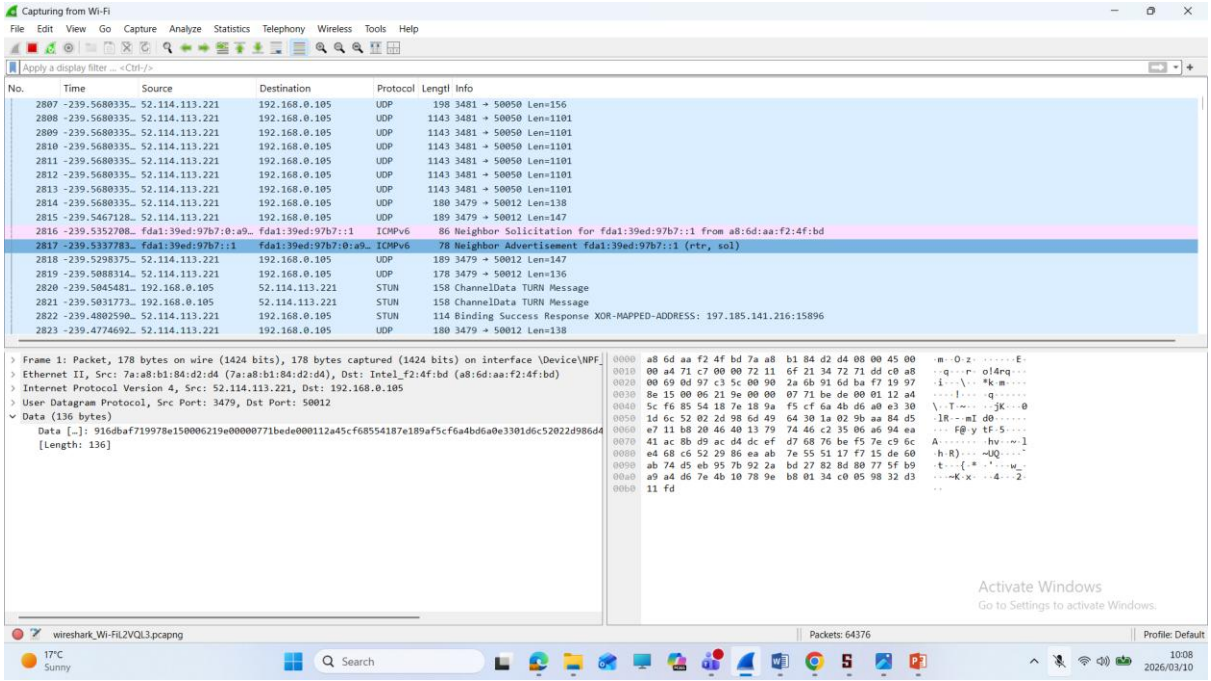
DNS



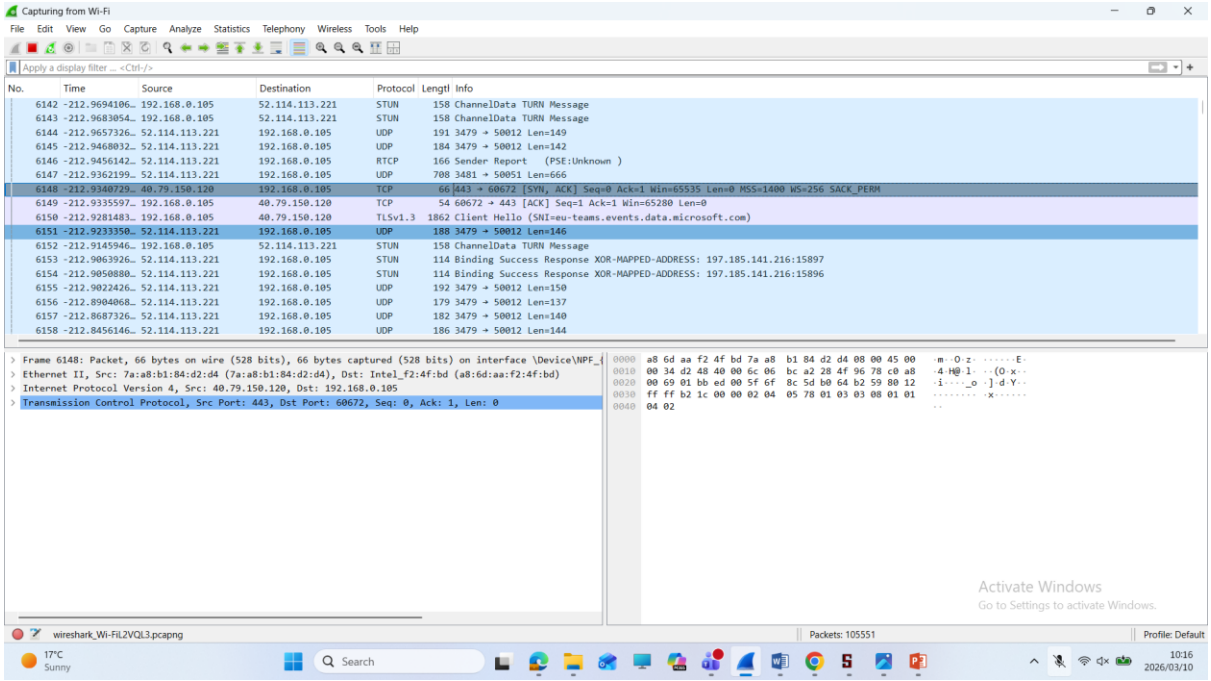
ARP



ICMP


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
2807 -239.5680335… 52.114.113.221 192.168.0.105 UDP 198 3481 → 50050 Len=156
2808 -239.5680335… 52.114.113.221 192.168.0.105 UDP 1143 3481 → 50050 Len=1101
2809 -239.5680335… 52.114.113.221 192.168.0.105 UDP 1143 3481 → 50050 Len=1101
2810 -239.5680335… 52.114.113.221 192.168.0.105 UDP 1143 3481 → 50050 Len=1101
2811 -239.5680335… 52.114.113.221 192.168.0.105 UDP 1143 3481 → 50050 Len=1101
2812 -239.5680335… 52.114.113.221 192.168.0.105 UDP 1143 3481 → 50050 Len=1101
2813 -239.5680335… 52.114.113.221 192.168.0.105 UDP 1143 3481 → 50050 Len=1101
2814 -239.5680335… 52.114.113.221 192.168.0.105 UDP 180 3479 → 50012 Len=138
2815 -239.5467128… 52.114.113.221 192.168.0.105 UDP 189 3479 → 50012 Len=147
2816 -239.5352708… fda1:39ed:97b7:0:a9… fda1:39ed:97b7::1 ICMPv6 86 Neighbor Solicitation for fda1:39ed:97b7::1 from a8:6d:aa:f2:4f:bd
2817 -239.5337783… fda1:39ed:97b7::1 fda1:39ed:97b7:0:a9… ICMPv6 78 Neighbor Advertisement fda1:39ed:97b7::1 (rtr, sol)
2818 -239.5298375… 52.114.113.221 192.168.0.105 UDP 189 3479 → 50012 Len=147
2819 -239.5088314… 52.114.113.221 192.168.0.105 UDP 178 3479 → 50012 Len=136
2820 -239.5045481… 192.168.0.105 52.114.113.221 STUN 158 ChannelData TURN Message
2821 -239.5031773… 192.168.0.105 52.114.113.221 STUN 158 ChannelData TURN Message
2822 -239.4802590… 52.114.113.221 192.168.0.105 STUN 114 Binding Success Response XOR-MAPPED-ADDRESS: 197.185.141.216:15896
2823 -239.4774692… 52.114.113.221 192.168.0.105 UDP 180 3479 → 50012 Len=138
Frame 1: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface \Device\NPF_
Ethernet II, Src: 7a:a8:b1:84:d2:d4 (7a:a8:b1:84:d2:d4), Dst: Intel_f2:4f:bd (a8:6d:aa:f2:4f:bd)
Internet Protocol Version 4, Src: 52.114.113.221, Dst: 192.168.0.105
User Datagram Protocol, Src Port: 3479, Dst Port: 50012
Data (136 bytes)
Data […]: 916dbaf719978e150006219e00000771bede000112a45cf68554187e189af5cf6a4bd6a0e3301d6c52022d986d4
[Length: 136]
0000 a8 6d aa f2 4f bd 7a a8 b1 84 d2 d4 08 00 45 00
0010 00 a4 71 c7 00 00 72 11 6f 21 34 72 71 dd c0 a8
0020 00 69 0d 97 c3 5c 00 90 2a 6b 91 6d ba f7 19 97
0030 8e 15 00 06 21 9e 00 00 07 71 be de 00 01 12 a4
0040 5c f6 85 54 18 7e 18 9a f5 cf 6a 4b d6 a0 e3 30
0050 1d 6c 52 02 2d 98 6d 49 64 30 1a 02 9b aa 84 d5
0060 e7 11 b8 20 46 40 13 79 74 46 c2 35 06 a6 94 ea
0070 41 ac 8b d9 ac d4 dc ef d7 68 76 be f5 7e c9 6c
0080 e4 68 c6 52 29 86 ea ab 7e 55 51 17 f7 15 de 60
0090 ab 74 d5 eb 95 7b 92 2a bd 27 82 8d 80 77 5f b9
00a0 a9 a4 d6 7e 4b 10 78 9e b8 01 34 c0 05 98 32 d3
00b0 11 fd
Activate Windows
Go to Settings to activate Windows.
wireshark_Wi-FiL2VQL3.pcapng
Packets: 64376
Profile: Default
17°C Sunny
Search
10:08 2026/03/10

TCP


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
6142 -212.9694106… 192.168.0.105 52.114.113.221 STUN 158 ChannelData TURN Message
6143 -212.9683054… 192.168.0.105 52.114.113.221 STUN 158 ChannelData TURN Message
6144 -212.9657326… 52.114.113.221 192.168.0.105 UDP 191 3479 → 50012 Len=149
6145 -212.9468032… 52.114.113.221 192.168.0.105 UDP 184 3479 → 50012 Len=142
6146 -212.9456142… 52.114.113.221 192.168.0.105 RTCP 166 Sender Report (PSE:Unknown)
6147 -212.9362199… 52.114.113.221 192.168.0.105 UDP 708 3481 → 50051 Len=666
6148 -212.9340729… 40.79.150.120 192.168.0.105 TCP 66 443 → 60672 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM
6149 -212.9335597… 192.168.0.105 40.79.150.120 TCP 54 60672 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
6150 -212.9281483… 192.168.0.105 40.79.150.120 TLSv1.3 1862 Client Hello (SNI=eu-teams.events.data.microsoft.com)
6151 -212.9233350… 52.114.113.221 192.168.0.105 UDP 188 3479 → 50012 Len=146
6152 -212.9145946… 192.168.0.105 52.114.113.221 STUN 158 ChannelData TURN Message
6153 -212.9063926… 52.114.113.221 192.168.0.105 STUN 114 Binding Success Response XOR-MAPPED-ADDRESS: 197.185.141.216:15897
6154 -212.9050880… 52.114.113.221 192.168.0.105 STUN 114 Binding Success Response XOR-MAPPED-ADDRESS: 197.185.141.216:15896
6155 -212.9022426… 52.114.113.221 192.168.0.105 UDP 192 3479 → 50012 Len=150
6156 -212.8904068… 52.114.113.221 192.168.0.105 UDP 179 3479 → 50012 Len=137
6157 -212.8687326… 52.114.113.221 192.168.0.105 UDP 182 3479 → 50012 Len=140
6158 -212.8456146… 52.114.113.221 192.168.0.105 UDP 186 3479 → 50012 Len=144
Frame 6148: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{
Ethernet II, Src: 7a:a8:b1:84:d2:d4 (7a:a8:b1:84:d2:d4), Dst: Intel_f2:4f:bd (a8:6d:aa:f2:4f:bd)
Internet Protocol Version 4, Src: 40.79.150.120, Dst: 192.168.0.105
Transmission Control Protocol, Src Port: 443, Dst Port: 60672, Seq: 0, Ack: 1, Len: 0
0000 a8 6d aa f2 4f bd 7a a8 b1 84 d2 d4 08 00 45 00
0010 00 34 d2 48 40 00 6c 06 bc a2 28 4f 96 78 c0 a8
0020 00 69 01 bb ed 00 5f 6f 8c 5d b0 64 b2 59 80 12
0030 ff ff b2 1c 00 00 02 04 05 78 01 03 03 08 01 01
0040 04 02
Activate Windows
Go to Settings to activate Windows.
wireshark_Wi-FiL2VQL3.pcapng
Packets: 105551
Profile: Default
17°C Sunny
Search
10:16 2026/03/10